Credit Risk Modelling Interpretation

1. Overview

This project builds a Probability of Default (PD) model using logistic regression for credit risk modelling and Loss given Default, Exposure at Default analysis for Score Card Analysis.

The model was trained on the training dataset and validated using two important credit risk performance metrics:

1) Kolmogorov–Smirnov (KS) statistic
2) Receiver Operating Characteristic (ROC) curve and Area Under the Curve (AUC).

These metrics are widely used in banking and credit risk modelling to evaluate how well the model separates good borrowers (non-defaulters) from bad borrowers (defaulters).

STEPS:

Project executing following steps,

1) Importing The Data and Explore the data

2) Data Cleaning by Remove Duplicates, Drop missing Values, Same data type for each column, Transforming Numerical Features to Categories, One Hot encoding Categorical Variables, Feature selection by Analyzing Weight of Evidence and Information value for each Feature

3) Machine Learning Algorithms to Predict Probability of Default, Loss given Default, Exposure at Default for Score Card Analysis

4) Model Evaluation for our prediction with Kolmogorov–Smirnov (KS) statistic Receiver Operating Characteristic (ROC) curve and Area Under the Curve (AUC).

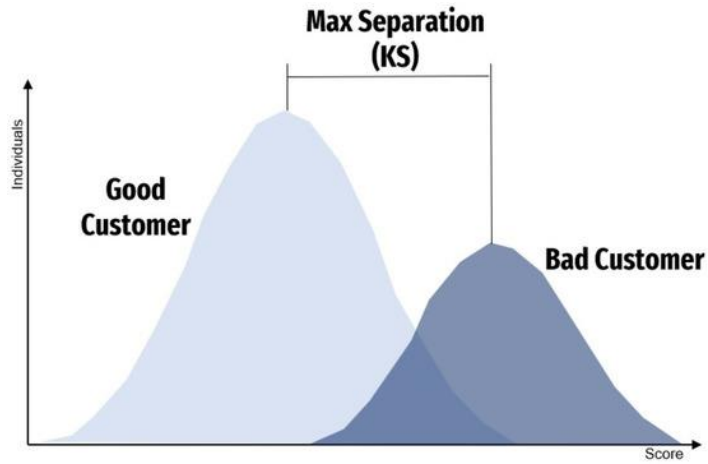
2. KS Statistic Interpretation

The Kolmogorov–Smirnov (KS) statistic measures the maximum difference between the cumulative distribution of good borrowers and bad borrowers.

$KS = \max |F_good(x) - F_bad(x)|$

In this project the model produced a KS value of approximately 0.899 (89.9%).

Interpretation scale:
KS < 0.20 : Weak model
0.20 – 0.40 : Acceptable model
0.40 – 0.60 : Good model
> 0.60 : Very strong model



A KS value of 0.899 indicates extremely strong discriminatory power.
The model is able to clearly separate high-risk borrowers from low-risk borrowers.

In the KS plot:
• The black curve represents cumulative bad borrowers (defaulters)
• The blue curve represents cumulative good borrowers (non-defaulters)
• The red dashed line indicates the maximum separation point between the two curves

This means the model identifies a large proportion of defaulters within a relatively small portion of the borrower population, which is desirable for credit risk screening.

3. ROC Curve Interpretation

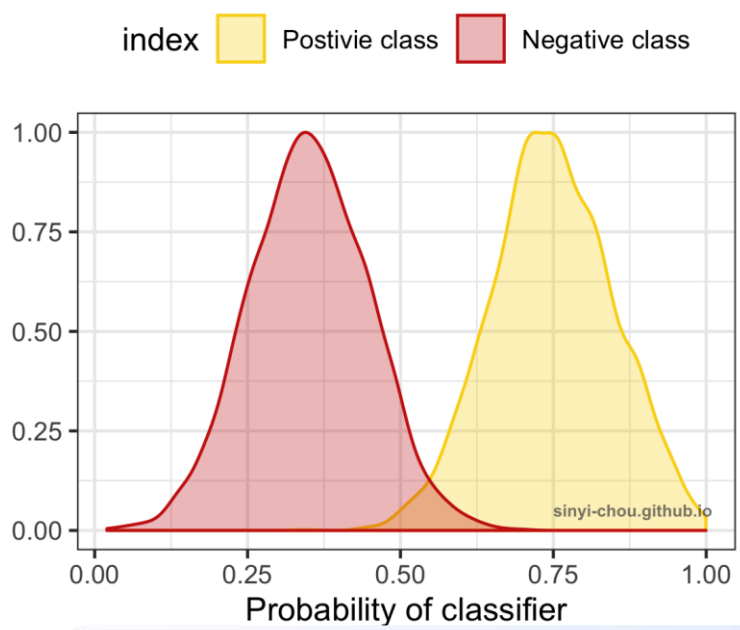
The ROC curve plots the relationship between:

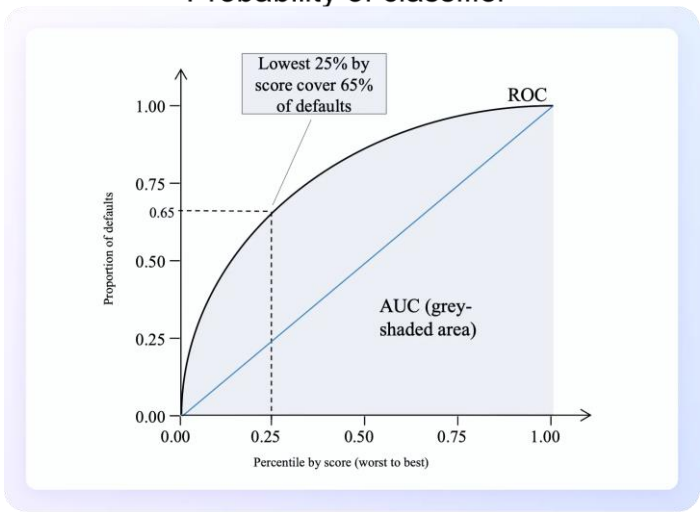
True Positive Rate (TPR) – correctly identified defaulters
False Positive Rate (FPR) – non-defaulters incorrectly classified as defaulters

The Area Under the ROC Curve (AUC) measures the overall classification ability of the model.

AUC Interpretation:
0.50 : Random model
0.60 – 0.70 : Weak
0.70 – 0.80 : Acceptable
0.80 – 0.90 : Good
> 0.90 : Excellent

In this project the AUC value is approximately 0.98.





This means the model has near-perfect classification performance.
If we randomly select one defaulter and one non-defaulter, the model will correctly rank the defaulter as riskier about 98% of the time.

4. Relationship Between KS and AUC

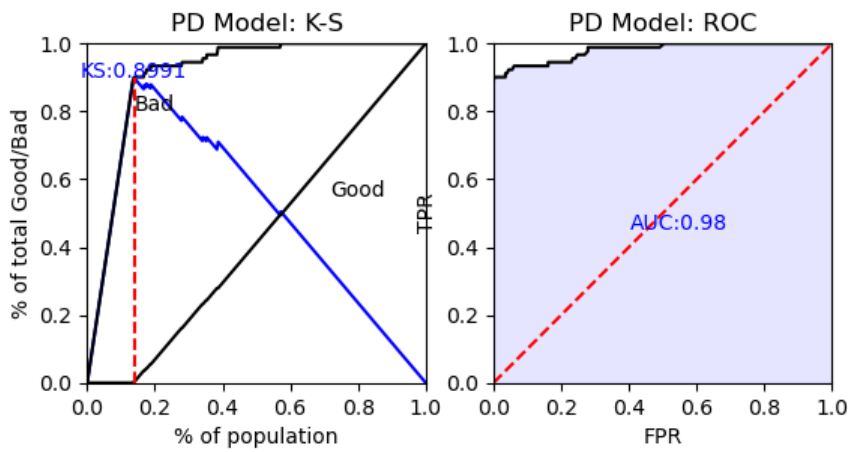
There is an approximate relationship between KS and AUC:

$KS \approx 2 \times AUC - 1$

Using AUC = 0.98:

$KS \approx 2(0.98) - 1 = 0.96$

This is consistent with the observed KS value (~0.899), confirming strong model discrimination.



5. Implications for Credit Risk Modelling

The PD model demonstrates very strong predictive ability in identifying borrowers who are more likely to default.

This allows lenders to:
- Identify high-risk borrowers
- Reduce exposure to risky credit applicants
- Implement risk-based pricing
- Improve portfolio risk management

The predicted Probability of Default (PD) can also be used with other credit risk parameters to calculate Expected Loss:

Expected Loss = PD × LGD × EAD

Where:
PD = Probability of Default
LGD = Loss Given Default
EAD = Exposure at Default

Credit Risk Score Card Output:

Borrower	PD	LGD	EAD	Expected Loss	Credit Score
0	0.0159	0.6	3871.69	36.96	686
1	0.0118	0.6	10021.05	70.91	707
2	0.9509	0.6	6739.86	3845.16	175
3	0.00001	0.6	161.24	0.00096	1218
4	0.0303	0.6	6333.39	114.98	637

Interpretation of Variables

I. Probability of Default (PD)

PD represents the **likelihood that a borrower will default on their loan**.

Borrower	PD Interpretation
0	Very low default risk (1.59%)
1	Very low default risk (1.17%)
2	Extremely high default risk (95.08%)
3	Almost zero default probability
4	Low default probability (3.03%)

Borrower **2** has the **highest probability of default**, indicating very high credit risk.

II. Loss Given Default (LGD)

LGD represents the **percentage of exposure the lender expects to lose if default occurs**.

This means the bank assumes **60% of the loan will be lost if the borrower defaults**.

This assumption is common in simplified credit risk models when **recovery data is not available**.

III. Exposure at Default (EAD)

EAD represents the **total outstanding loan amount at the time of default**.

Borrower	EAD Meaning
1	Highest loan exposure (~10021)
3	Very small loan exposure (~161)
2	Large exposure (~6739)

IV. Expected Loss (EL)

Expected Loss is calculated using the **Basel credit risk formula**:

$$Expected\ Loss = PD \times LGD \times EAD$$

This represents the **average amount the lender expects to lose from each borrower**.

Interpretation

Borrower	Expected Loss	Meaning
0	36.96	Small expected loss
1	70.91	Moderate loss risk
2	3845.16	Extremely high loss risk
3	0.00096	Almost zero expected loss
4	114.98	Moderate risk

Borrower **2 dominates portfolio risk**, contributing the **largest expected loss**.

V. Credit Score Interpretation

The credit score is derived from the predicted PD.

Higher score → **lower risk**.

Score Range	Risk Level

Score Range	Risk Level
< 400	Very High Risk
400 – 600	High Risk
600 – 700	Moderate Risk
> 700	Low Risk

Borrower Risk Segmentation:

Borrower	Score	Risk Category
0	**686**	**Moderate Risk**
1	**707**	**Low Risk**
2	**175**	**Very High Risk**
3	**1218**	**Extremely Low Risk**
4	**637**	**Moderate Risk**

Borrower **2 has an extremely low score**, indicating severe credit risk.

Borrower **3 has a very high score**, suggesting a very safe borrower.

VI. Portfolio Risk Insights

From this sample:

High Risk Borrower

Borrower 2

- PD ≈ 95%
- Expected loss ≈ 3845
- Score ≈ 175

This borrower should likely be:

- rejected
- charged higher interest
- given lower credit limits

Low Risk Borrowers

Borrowers 0, 1, and 3 show **low expected losses and high scores**.

These borrowers represent **good lending opportunities**.

VII. Economic Interpretation for the Bank

The scorecard allows the bank to:

1. **Rank borrowers by risk**
2. **Estimate expected portfolio losses**
3. **Make lending decisions**

Example policy:

Score	Lending Decision
>700	Approve loan
600–700	Approve with caution
<600	Reject or price higher

Using this rule:

- Borrower **2 would be rejected**
- Borrowers **0,1,4 approved with monitoring**
- Borrower **3 strongly approved**

VIII. Key Insight from the Model

The scorecard successfully links:

- **PD (risk probability)**
- **EAD (loan exposure)**
- **LGD (loss severity)**

to estimate **Expected Loss**, which is the **core concept in Basel credit risk regulation**.

6. Important Note

The extremely high AUC (0.98) and KS (0.899) values suggest that the dataset may be relatively simple or small, which can sometimes lead to very high-performance metrics.

In real banking datasets typical values are:
AUC: 0.70 – 0.85
KS: 0.30 – 0.50

Hence the model performs excellently for this dataset but further validation on larger and more complex datasets would be required in real financial applications.

7. Conclusion

The logistic regression PD model demonstrates excellent discriminatory power and predictive accuracy.
The KS statistic and ROC curve both confirm that the model is highly effective in distinguishing between defaulters and non-defaulters.

These results indicate that the model can be used as a reliable credit scoring tool for estimating borrower risk and supporting lending decisions.